

Day 1



90 Days SDET
Interview Prep

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90 Days SDET Interview Prep

Day 1

Explain the OOP Concepts and where you have used it in your Automation Project?

1. ENCAPSULATION

- ✓ Encapsulation is the binding of data (instance variables) and methods (getters and setter) together.
- ✓ It mainly focuses on restricting direct access to instance variables from outside the class.
- ✓ Instance variables are marked as private and are accessed only through getters and setters.
- ✓ This helps protect instance variables from illegal or unintended assignments.

Implementation in Framework

In the framework, all classes that contain instance variables mark them as private, and getters and setters are provided to access or update them.

This is commonly used in:

- POJO classes
- BrowserUtility
- Other utility and page classes

Why it is used:

- All fields are private
- Only getters are provided (immutable object pattern where applicable)
- Data can be set only through constructors
- Prevents unintended modification after object creation

2. INHERITANCE

- ✓ Inheritance allows a class to inherit properties and methods from another class, which helps in code reuse and maintainability.

Inheritance Hierarchy in Framework

BrowserUtility (Parent/Base Class)

↓

- ├─ HomePage (Child)
- ├─ MyAccountPage (Child)
- ├─ AddressPage (Child)
- └─ ShoppingCartPage (Child)

Benefits:

- ✓ All page classes inherit common browser operations
- ✓ No code duplication across page classes
- ✓ Single point of maintenance for browser-related operations
- ✓ Consistent behavior across all pages

3. POLYMORPHISM

- ✓ Polymorphism allows objects to be treated as instances of their parent class, and methods to behave differently based on the object type.

- **Method Overloading (Compile-time Polymorphism)**

- ✓ Benefits of Method Overloading:
- ✓ Same method name with different parameters
- ✓ Flexibility in how methods are called
- ✓ More intuitive and readable APIs
- ✓ Method selection happens at compile time

- **Where You have used Method Overloading?**

`BrowserUtility Class`

- **Interface Implementation (Runtime Polymorphism)**

`TestListener implements ITestListener`

Benefits:

- ✓ TestNG framework treats TestListener as an ITestListener type
- ✓ Actual implementation is decided at runtime
- ✓ Enables loose coupling with the TestNG framework
- ✓ Allows multiple implementations of the same interface

4. ABSTRACTION

Abstraction hides complex implementation details and exposes only essential features to the user.

Implementation in Framework

Utility classes act as an abstraction layer:

- ✓ BrowserUtility acts as a wrapper over Selenium
- ✓ JSONUtility abstracts JSON parsing
- ✓ CSVReaderUtility abstracts OpenCSV library usage
- ✓ Page Object Model (POM) follows the abstraction pattern

Benefits of Abstraction:

- ✓ Test cases are readable and easy to maintain
- ✓ Implementation changes do not affect test code
- ✓ Reduces complexity for test writers
- ✓ Clearly separates what to do from how to do it

5. COMPOSITION (HAS-A Relationship)

Composition represents a HAS-A relationship, where a class contains objects of other classes as its members.

Implementation in Framework

`BrowserUtility` has:

- `WebDriver`
- `Logger`
- `WebDriverWait`

Composition Relationships:

- `BrowserUtility` HAS-A `WebDriver`
- `BrowserUtility` HAS-A `Logger`
- `BrowserUtility` HAS-A `WebDriverWait`

6. ASSOCIATION

Association represents a relationship between two independent classes through their objects.

Implementation in Framework

One-to-one association is used between page classes.

Example:

HomePage → LoginPage

Benefits:

- Models real user navigation flow
- Pages clearly know which pages they can navigate to
- Loose coupling, as pages remain independent classes

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